CONFOEDERATIO,
RESEARCH DIVISION (CRD).

POLOTSK, TUROV AND THE BELARUSIAN LANDS

Population and Towns of the Dvina, Pripyat and Neman Basins

8 November 2026 - Aust Kätzchen | austkaeztchen@confoederatio.org

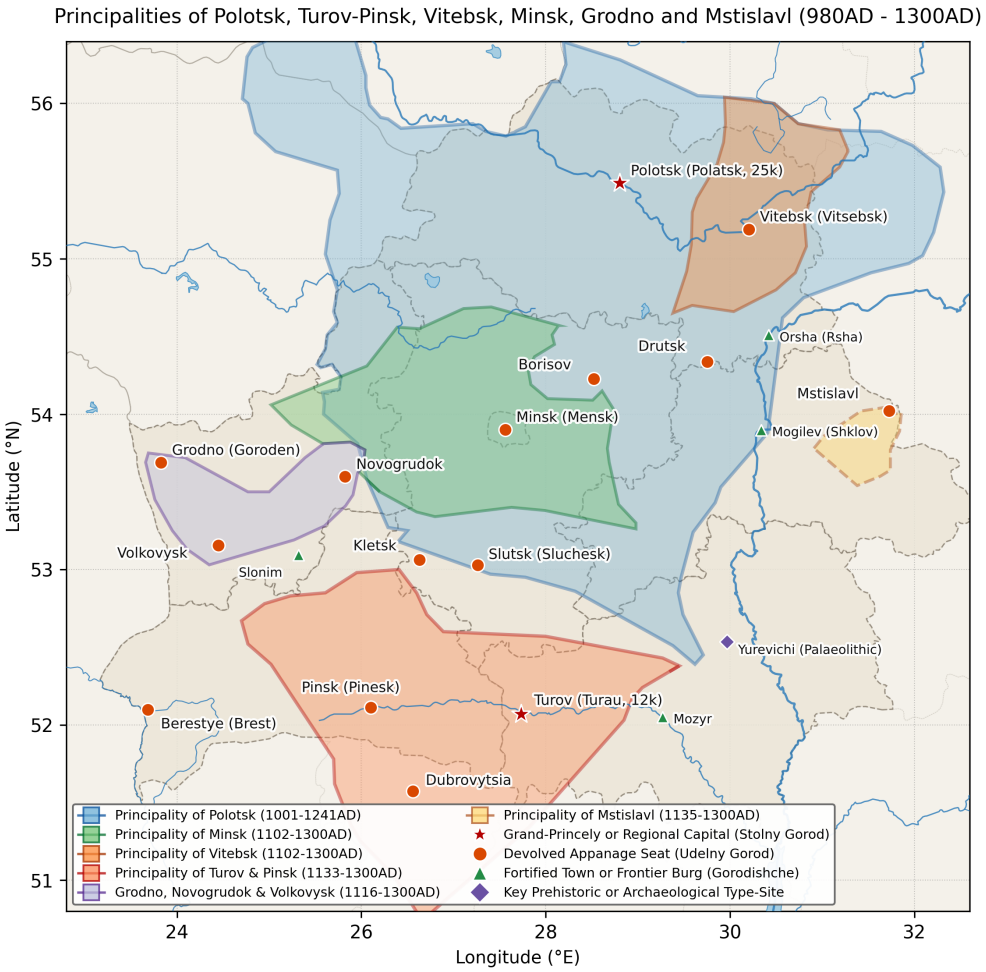


Figure 1. Approximate boundaries of the Belarusian principalities (207,6k km^2), showing the Izyaslavichi capital at Polotsk on the Western Dvina (25k in 1200AD), the Polesian seats at Turov (12k), Pinsk and Berestye, the Upper Neman towns of Black Ruthenia (Grodno and Novogrudok) and the Dvina vassal forts of Jersika and Koknese.

Abstract

Straddling the river portages that linked the Dnieper with the Baltic Sea along the Western Dvina, Neman and Western Bug, the lands of Polotsk, Turov and Belarusian Rus (207,6k km^2) formed the north-western wing of the East Slavic world [1][2][3]. From a Final Palaeolithic base of 8k reindeer hunters in 10000BC and 32k Neolithic lake-dwellers in 5000BC, regional population rose to 240k in 500BC across more than

1,1k Iron Age hillforts and reached 340k by 500AD [4][5]. Between the sixth and tenth centuries, Slavic settlement by the Krivichi, Dregovichi and Radimichi merged with local Baltic communities to carry the population from 460k in 800AD to 620k in 1000AD [6].

Under the independent Izyaslavichi princes of Polotsk and the Rurikid rulers of Turov, Pinsk and Grodno, the region reached a pre-Mongol peak of 980k inhabitants in 1200AD across 188 fortified towns (*gorodishcha*), led by Polotsk (25k), Turov (12k), Vitebsk (9,5k), Minsk (9k), Grodno (8,5k), Pinsk (8k), Novogrudok (7,5k) and Berestye (7,5k) [1][7][8]. During Batu Khan's campaigns of 1239-1241, Mongol columns sacked Gomel on the lower Sozh and raided the southern Pripyat towns of Turov, Mozyr and Berestye, but the dense Polesian marshes shielded Polotsk, Vitebsk, Minsk and Black Ruthenia from attack. Total population dipped by only 16,3% to 820k in 1250AD and recovered to 860k by 1300AD, providing the urban and military backbone of the early Grand Duchy of Lithuania [9][10]. These figures connect directly with the neighbouring Russian series for Novgorod-Pskov (310k in 1000AD and 480k in 1200AD) and Smolensk east of 31°E (220k in 1000AD and 340k in 1200AD).

Background

Across the Kievan Rus and appanage periods (980AD - 1300AD), North-Western Rus fell into five main river-basin groups [1][2][3][8].

In the north along the Western Dvina (Daugava), the senior Izyaslavichi patrimony comprised the Principality of Polotsk (1001-1241AD), the Principality of Vitebsk (1102-1321AD) with *Orsha* and *Kopys*, and the lower Dvina tributary principalities of Jersika (1001-1210AD) and Koknese (1191-1210AD) in Latgale and Selonia [7][11][12][13]. Across the central Minsk Upland and the Upper Berezina, Svisloch and Drut portages lay the southern Izyaslavichi principalities of Minsk (1102-1327AD, moved from the tenth- and eleventh-century *Menka* fort to the Svisloch-Nemiga *Zamchishche* after 1067AD), Drutsk (1102-1251AD) and Borisov (1102-1301AD), alongside *Zaslavl* (*Izyaslavl*), *Logoisk* (*Logozhsk*) and *Svisloch-on-Berezina* [1][14]. In the south along the Pripyat, Pina, Yaselda, Sluch, Lan, Ptich and Horyn rivers, Pripyat Polesia and the Berestye Land encompassed the principalities of Turov (1133-1341AD), Pinsk (1175-1321AD), Kletsk (1148-1401AD) with *Slutsk* and *Kopyl*, and Dubrovytsia (1175-1301AD), together with the Western Bug-Mukhavets frontier towns of Berestye (*Brest*) and Kamenets [3][15][16]. To the west along the Upper Neman, Shchara and Rossa rivers, Black Ruthenia (*Chornaya Rus*) covered the Slavic-Baltic borderland around the principalities of Grodno (*Goroden*, 1128-1201AD), Novogrudok (1201-1241AD) and Volkovysk (1201-1241AD), with *Slonim*, *Zditovo-on-Molchad* and *Turiisk-on-Neman* [9][17][18]. In the east along the Vikhra, Upper Dnieper and Sozh rivers west of 31°E stood the Principality of Mstislavl (1181-1198AD) and the western Smolensk and northern Chernigov-Radimichi towns of *Propoisk* (*Prupoi*), *Krichev*, *Luchin*, *Shklov*, *Mogilev*, *Radoml*, *Chechersk*, *Rechitsa* and *Gomel* (*Homiy*) [19][20].

Date	Population	Low	High	Density (per km²)	Urban Pop.	Historical Notes
10000BC	8k	4,5k	13k	0,04	-	Yurevichi, Berdizh, Swiderian & Grensk Final Palaeolithic
5000BC	32k	20k	48k	0,15	-	Narva, Neman, Dnieper-Donets & Upper Dnieper Neolithic
2500BC	78k	52k	110k	0,38	-	Krivina pile dwellings (Asavets), Middle Dnieper & Corded Ware

1200BC	135k	95k	185k	0,65	-	Trzciniec-Sosnytsia & North Belarusian Bronze Age
500BC	240k	175k	315k	1,16	-	Milograd, Brushed Pottery & Dnieper-Dvina (>1,1k hillforts)
200AD	295k	220k	380k	1,42	-	Polesian Zarubintsy (Chaplin), Late Brushed Pottery & Wielbark
500AD	340k	255k	435k	1,64	12k	Bancerovshchina-Tushemlya, Kolochin & Prague-Korchak
800AD	460k	350k	580k	2,22	35k	Polotsk Krivichi Long Barrows, Dregovichi & Radimichi
900AD	530k	410k	660k	2,55	52k	Early Polotsk (Gorodishche), Vitebsk, Turov & Menka crafts
1000AD	620k	490k	760k	2,99	78k	Rurikid & Izyaslavichi consolidation in Polotsk & Svyatopolk in Turov
1100AD	790k	640k	950k	3,81	118k	Apogee of Vseslav the Sorcerer (1044-1101AD) & Turov Yuryevichi
1200AD	980k	850k	1,12M	4,72	162k	Pre-Mongol urban & kurgan peak across 188 gorodishcha
1240AD	940k	810k	1,08M	4,53	152k	Batu's southern raids on Gomel (1239), Turov, Mozyr & Berestye (1240-1241)
1250AD	820k	690k	960k	3,95	135k	Post-1240 trough in Polesia & Sozh; Polotsk, Vitebsk, Minsk & Novogrudok unstormed
1300AD	860k	720k	1M	4,14	148k	Early Lithuanian-Ruthenian realm (Black Ruthenia, Polotsk & Vitebsk)

Table 1. Estimated population, uncertainty ranges, settlement density and urban totals across the Western Dvina, Pripyat, Neman and Sozh basins (207,6k km^2) from the Final Palaeolithic camps of Yurevichi and Berdizh (10000BC) to the early Lithuanian-Ruthenian realm (1300AD).

Results

Prehistory and the Iron Age

Final Palaeolithic and Mesolithic

Human settlement across the Belarusian river basins began during the warmer intervals of the Late Pleistocene before the Last Glacial Maximum (26000BC - 21000BC). The two best-known Upper Palaeolithic mammoth-hunter camps in Belarus are Yurevichi (Yuravichy, Kalinkovichi district) on a high terrace of the lower Pripyat (c. 24500BC) and Berdizh (Berdydzh, Chechersk district) on the Sozh (c. 21500BC), where sunken dwellings framed with mammoth skulls, tusks and long bones were excavated by K. M. Polikarpovich and V. D. Budko [2][4]. After the Poozerie (Valdai) ice sheet retreated from northern Vitebsk around 12000BC - 10000BC, Final Palaeolithic reindeer-hunting bands of the Swiderian, Lyngby and Grensk cultures moved onto the sandy plains and lake shores of the Neman, Pripyat, Upper Dnieper, Sozh and Western Dvina valleys (*Grensk* and *Korma* on the Sozh, *Ankovichi* in the Poozerie and *Borok* on the Neman). At 10000BC, the population across this tundra-parkland zone is estimated at 8k inhabitants (range 4,5k - 13k, 0,04 per km^2) [4].

During the Boreal and Atlantic Mesolithic (8000BC - 5000BC), warmer temperatures and the spread of pine-birch and oak-hornbeam forests supported larger hunter-fisher communities across three cultural zones: the Kunda culture in the northern lake district (Western Dvina basin), the Janislawice (Neman) Mesolithic across the Neman and Western Bug basins and the Sozh and Upper Dnieper Mesolithic in eastern Belarus [4]. Over 400 Mesolithic camps have been recorded along lake shores and river junctions.

Neolithic River and Lake Settlements

The Neolithic period (5000BC - 2200BC) brought a four-fold increase in population (32k inhabitants at 5000BC, rising to 78k by 2500BC), supported by lake fishing, wildfowling and the gradual spread of stockbreeding and millet, barley and emmer cultivation [2][4]. In the Vitebsk lake district, the Narva and Usvyaty cultures built waterlogged pile-dwelling villages in the Krivina peat basin (Beshenkovichi and Senno districts: *Asavets 2*, *Asavets 7*, *Kryvina 1* and *Aztsy*), excavated by M. M. Charniauski; waterlogged layers at *Asavets* preserved wooden fish-weirs, dugout canoes, horn hoes, Baltic amber ornaments and carved bone and wooden figures [4]. Across the Upper Neman and Western Bug, the Neman Culture made pointed-bottom and flat-bottom pots tempered with plant fibres and crushed granite (*Kamen*, *Rusakovo* and *Dobry Bor*), and worked the underground Krasnoselsky flint mines on the Rossa river near Volkovysk, where over 1k vertical shafts up to 6 - 9 m deep were dug into chalk strata with antler picks to extract dark flint for polished axes traded across the Baltic-Dnieper region [4]. At the same time, the Dnieper-Donets Culture (in Pripyat Polesia and the Lower Sozh) and the Upper Dnieper Culture (in eastern Belarus) lived in permanent riverside villages along the Pripyat, Yaselda, Berezina and Sozh (*Litvinovichi*, *Strumen* and *Grodno-Neman*).

Bronze and Iron Age Hillforts

During the Late Eneolithic and Bronze Age (2500BC - 700BC), pastoral and farming communities of the Middle Dnieper Culture (2400BC - 1500BC, a branch of the Corded Ware or Battle-Axe horizon, with major barrow and flat cemeteries at *Strelitsa* and *Khodosovichi* near Rogachev), Polesian and Strzyzow groups and the North Belarusian and Trzciniec-Sosnytsia cultures (1500BC - 1000BC) cleared river terraces for cattle, pigs, sheep and goats alongside bronze-casting workshops using imported Carpathian and Alpine copper [4]. The regional population grew from 78k at 2500BC to 135k at 1200BC.

The Early and Middle Iron Age (700BC - 400AD) introduced local bog-iron smelting and covered the Belarusian landscape with more than 1,1k fortified timber-and-earth hillforts (*gorodishcha*).

Archaeologists divide these hillforts into three east-west belts that foreshadow early medieval Baltic and Slavic settlement [5][21][22]. In northern Belarus and the Vitebsk lake district, the Dnieper-Dvina Culture (700BC - 400AD) occupied the Western Dvina and upper Dnieper basins (*Polotsk-Gorodishche*, *Vitebsk-Zamkovaya Gora*, *Lukoml* and *Novye Bateki*), building small cliff-top hillforts (0,3 - 1,2 ha) with concentric earthen ramparts and timber palisades and living by slash-and-burn (*podseka*) farming, stockbreeding and ironworking [5]. Across central and north-western Belarus, the Brushed Pottery Culture (*Kultura Shtrikhovannoy Keramiki*, 700BC - 400AD) held the Minsk Upland and Upper Neman basin (*Malyshki*, *Zazony*, *Viazynka*, *Novogrudok* and *Slonim*), where A. G. Mitrofanov excavated multi-room post-built longhouses inside ring-forts [21]. In southern Belarus across the Pripyat and Lower Berezina-Sozh valleys, the Milograd Culture (700BC - 100BC) and Zarubintsy Culture (200BC - 200AD) spanned Gomel, southern Mogilev and Brest Polesia (*Milograd*, *Goroshkov*, *Chaplin*, *Mokhov*, *Voronino* and *Otverzhichi*); unlike the small northern hillforts, Milograd and Polesian Zarubintsy communities built larger lowland forts (2 - 8 ha) and open farming villages (3 - 10 ha) equipped with iron plough-tips (*ralniki*), sickles and rotary querns [5][22].

By 500BC, more than 1,1k hillforts and open villages (*selyshcha*) supported 240k inhabitants (range 175k - 315k), rising to 295k by 200AD during the Zarubintsy and Wielbark period. Across the Late Roman and Migration periods (200AD - 500AD), contact between early Slavic farmers and local Baltic communities produced the Kiev, Kolochin (in the Sozh basin, named after the *Kolochin* hillfort near Rechitsa), Bancerovshchina-Tushemlya (central and northern Belarus, named after the *Bancerovshchina* hillfort on the Svisloch near Minsk) and Prague-Korchak (Pripyat Polesia, at *Snyadin*, *Petrikov* and *Bereztsy*) cultures, bringing the 500AD regional population to 340k (range 255k - 435k) [5][6].

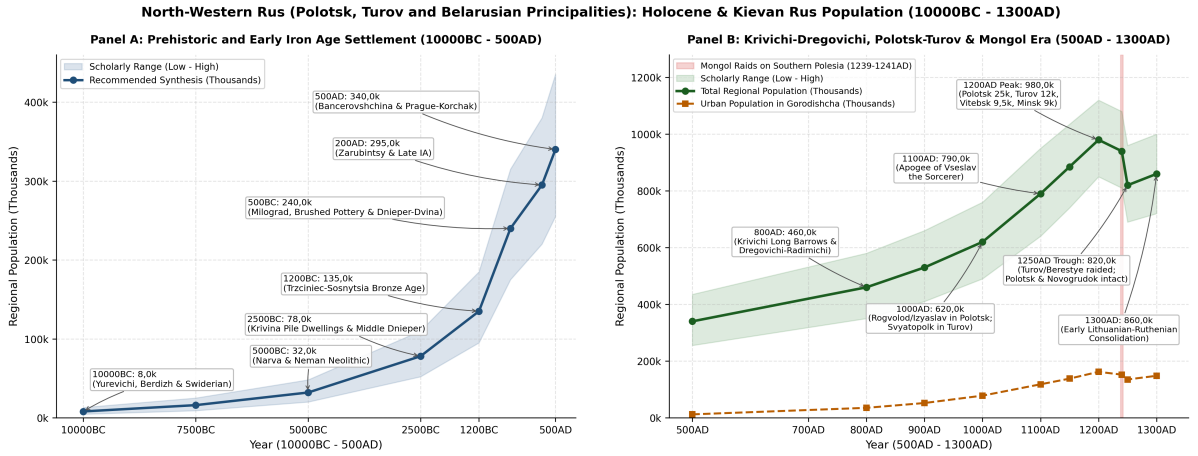


Figure 2. Shaded bands indicate uncertainty ranges around the central estimate for the 207,6k km^2 Belarusian river basins. Pre-Slavic settlement expanded across more than 1,1k Iron Age hillforts (240k at 500BC) and the Roman-era Zarubintsy and Wielbark horizons (295k at 200AD); medieval growth peaked at 980k in 1200AD before dipping by 16,3% to 820k in 1250AD and recovering to 860k by 1300AD.

Early Slavic and Kievan Rus Principalities

Tribal Confederations

Between the sixth and tenth centuries, the *Primary Chronicle* (*Povest Vremennykh Let*) and barrow (*kurgan*) excavations trace the growth of three major East Slavic tribal unions across Belarus, which gradually absorbed the local Baltic (East Lithuanian, Yotvingian and Latgalian-Selonian) population [2][6][20]. The Polotsk-Smolensk Krivichi (*Polochane*) settled along the Western Dvina, Polota, Drissa, Ulla, Kasplya and Upper Berezina rivers, leaving elongated Long Barrows (*Dlinnye Kurgany*, sixth to ninth centuries) with cremation burials, followed in the ninth to eleventh centuries by round barrows with bronze bracelet-shaped temple rings (*brasletoobraznye visochnye koltsa*) [1][6]. The Dregovichi (*Drygavichy*) occupied the Pripyat basin north to the Minsk Upland (meeting the Krivichi along the Svisloch-Berezina watershed at *Zaslavl*, *Minsk*, *Logoisk* and *Borisov*), where their burial mounds contain large granulated silver and bronze bead temple rings (*krupnozernistye businnye koltsa*) and timber burial chambers (*domoviny*) [15][20]. The Radimichi settled along the Sozh, Pronya, Iput, Besed and Ostyor rivers in eastern Belarus (with major hillforts at *Radoml*, *Propoisk* or *Prupoi* and *Gomel* or *Homiy*), marked by seven-rayed bronze temple rings (*semiluchevye visochnye koltsa*) [19][20]. Along the western and north-western edges, Yotvingians (*Yatvyagi*), Lithuanians (*Litva*), Latgalians and Selonians held the Upper Neman (*Grodno*, *Novogrudok*, *Volkovysk*, *Slonim*, *Lida* and *Krevo*) and Lower Western Dvina (*Braslav*, *Jersika*, *Koknese* and *Asote*) borderlands, where mixed Slavic-Baltic stone-ring barrows (*kamennye mogilniki*) mark a shared frontier zone [9][13][18].

The regional population grew from 460k in 800AD to 530k in 900AD and 620k in 1000AD, alongside 310k in *Novgorod-Pskov* and 220k in *Eastern Smolensk* within European Russia.

The Polotsk Land

Unlike the southern and eastern principalities of Kievan Rus, the Principality of Polotsk (1001-1241AD, 55,49°N, 28,8°E) kept its own ruling house under the Izyaslavichi (*Rogvolodovichi*), descendants of Prince Rogvolod (d. 978AD), his daughter Rogneda and her son by Vladimir the Great, Izyaslav Vladimirovich (d. 1001AD) [11][12]. Under Bryachislav Izyaslavich (1003-1044AD) and Vseslav Bryachislavich the Sorcerer (1044-1101AD), Polotsk controlled the entire Western Dvina trade route from the Vitebsk-Smolensk portages down to the Gulf of Riga, building the seven-domed Cathedral of St Sophia in Polotsk (c. 1050s) as a rival to the cathedrals of Kiev and Novgorod [7][11].

Excavations in Polotsk (*Polatsk*) by G. V. Shtykhov (1975, 1992) and S. V. Tarasov (1998) uncover one of the largest cities of pre-Mongol Eastern Europe [7][11]. The town grew from an early hillfort (*Gorodishche*, ninth and tenth centuries, 2,5 ha) on the Polota river, moved in the late tenth century to the Upper Castle (*Verkhny Zamok* or *Detinets*, 11 ha) at the Polota-Dvina junction around St Sophia Cathedral and the princely court, and spread across the Great Posad (*Velikiy Posad*, eleventh and twelfth centuries), the Lower Castle (*Nizhny Zamok* or *Zapadnaya Dvina Posad*, 63 ha), Zapolotye (west of the Polota), Ostrov (the island quarter), the Euphrosyne Suburb (around the 1128AD Saviour-Euphrosyne Monastery founded by St Euphrosyne of Polotsk, with the frescoed Church of the Transfiguration built by Master Ioann) and the Belchitsy princely seat on the left bank of the Dvina. By 1200AD, the built-up area of Polotsk covered 180 - 200 ha (190 ha in our estimates), housing 22k - 28k inhabitants (25k in our tables) across 3,8k - 4,2k urban homesteads (*dvory*), ringed by smaller towns at Lukoml (3,5k), Logoisk (*Logozhsk*, 3,5k), Zaslavl (*Izyaslavl*, 20 ha, 4k), Braslav (*Bryachislavl*, 3k), Usvyaty (2,8k) and Svisloch-on-Berezina (2,5k) [1][7][8].

After the death of Vseslav the Sorcerer in 1101AD, the Polotsk Land split into six smaller principalities (*udely*), each fostering local town growth. The Principality of Minsk (1102-1327AD, 53,9°N, 27,56°E) was founded by Gleb Vseslavich (d. 1119AD); E. M. Zagorulsky's excavations showed that early *Mensk* began in the tenth and eleventh centuries on the Menka river (at Gorodishche village, 16 km west of modern Minsk, with a 1,5 ha two-part fort and a 40 ha open settlement peaking at 2,4k inhabitants in 1000AD).

After the 1067AD Battle on the Nemiga River recorded in the *Primary Chronicle* and the *Tale of Igor's Campaign*, the fortress moved to the Minsk Zamchishche at the Svisloch-Nemiga junction (a 3 ha timber-laced rampart enclosing an unfinished eleventh-century stone church, surrounded by 35 ha of waterlogged *posad* log streets), reaching 38 ha and 9k inhabitants (8k - 10,5k) at 1200AD [2][14]. The Principality of Vitebsk (1102-1321AD, 55,19°N, 30,2°E), held by Svyatoslav and Davyd Vseslavich at the Vitba-Western Dvina junction, covered the *Verkhny Zamok* (Zamkovaya Gora), *Nizhny Zamok* and *Uzgorier* suburbs (50 ha in all), contained the 1140s Church of the Annunciation (built of alternating brick and dressed dolomite blocks), yielded the late thirteenth-century birch-bark letter of Stepan and held 9,5k inhabitants (8k - 11k) at 1200AD, alongside the Dnieper portage fortress of Orsha (*Rsha*, 1067AD, 22 ha, 4k inhabitants) and Kopys (1059AD, 2,8k) [1][12].

In the central portage country, the Principality of Drutsk (1102-1251AD, 54,34°N, 29,75°E) was ruled by Rogvolod Borisovich (fl. 1144-1171AD, patron of the 1178AD *Drutsk Gospel* and the inscribed *Boris Stones* on the Dvina), where L. V. Alekseev excavated a 2,5 ha citadel, a 9,5 ha *okolny gorod* and 23 ha of suburbs (35 ha in all, 4,5k inhabitants) [12]. The Principality of Borisov (1102-1301AD, 54,23°N, 28,52°E) was founded on the Berezina in 1102AD by Prince Boris Vseslavich at Stary Borisov (18 ha, 3,5k inhabitants) [1]. Downriver in Latgale and Selonia, the tributary principalities of Jersika (1001-1210AD, 56,25°N, 26,19°E) and Koknese (1191-1210AD, 56,64°N, 25,42°E) guarded the lower Western Dvina; according to Henry of Livonia's *Chronicon Livoniae*, Orthodox Prince Visvaldis (Vsevolod) of Jersika ruled from the 1,5 ha cliff-top hillfort and 12 ha suburb of *Jersika (Gerzike)* (2,5k inhabitants) until Bishop Albert of Riga and the Swordbrothers stormed and burned its Orthodox churches in 1209AD, while Prince Vyachko (Vetseke) of Koknese held the Dvina-Perse promontory fortress of *Koknese (Kukenois)* (10 ha, 2k inhabitants) until burning his own castle in 1208AD [12][13].

Turov-Pinsk and Berestye

Across southern Belarus, the Pripyat river and its ten navigable tributaries formed the Land of Turov and Pinsk. First mentioned under Prince Tury in 980AD and held in the eleventh century by the senior heirs to the Kievan throne (Svyatopolk Vladimirovich, Izyaslav Yaroslavich and Svyatopolk Izyaslavich), Turov (1133-1341AD, 52,07°N, 27,74°E) became an independent dynastic seat under Vyacheslav Vladimirovich and Yuri Yaropolkovich (d. c. 1167AD) after withstanding a ten-week siege in 1158AD [3][15].

P. F. Lysenko's excavations (1961-1968, 1992-2003) at Turov (*Turau*) on the Pripyat uncovered the 2,5 ha *Zamkova Hora* citadel (containing the brick foundations of the 1170s Cathedral of the Dormition, 29,3 x 17,9 m, seat of the preacher Bishop Cyril of Turov, c. 1130-1182AD, together with four lead patriarchal and episcopal seals and carved Ovruch slate sarcophagi), the 12,5 ha *Okolny Gorod* and 30 ha of riverside suburbs (45 ha in all, 12k inhabitants at 1200AD), flanked by Mozyr (16,5 ha, 3,5k), David-Gorodok (14 ha, 3k), Rogachev (14 ha, 3k), Bragin (11,5 ha, 2,8k) and Strelsk (1,8k) [3].

In the late twelfth century, Yuri Yaropolkovich's sons divided the Polesian basin into several smaller principalities. The Principality of Pinsk (1175-1321AD, 52,11°N, 26,1°E) stood at the Pina-Strumen-Pripyat junction (*Pinsk* or *Pinesk*, 1097AD), where a 3,2 ha citadel, 10,8 ha *okolny gorod* and 24 ha suburb (38 ha in all) held 8k inhabitants (6,5k - 9,5k) at 1200AD, alongside *Nobel* (1,6k) and *Pogost* (1,4k) [15]. The Principality of Kletsk (1148-1401AD, 53,06°N, 26,63°E) and Slutsk (*Sluchesk*, 1116AD, 53,03°N, 27,56°E) occupied the fertile loess ridge along the northern edge of Polesia, where Slutsk on the Sluch (25 ha, 5k inhabitants), Kletsk on the Lan (18 ha, 3,5k) and Kopyl (10 ha, 2,5k) were busy market and craft towns [15]. South of the Pripyat, the Principality of Dubrovytsia (1175-1301AD, 51,57°N, 26,56°E) on the Horyn river (18 ha, 3,5k inhabitants) was the seat of Prince Gleb Yuryevich (fl. 1183-1190AD, who fought at the Kayala in 1185AD), linked with Stepan (2,2k) and Vysotsk (1,25k) [15]. On the western edge of Polesia at the Western Bug-Mukhavets junction, the Berestye Land (*Beresteyskaya Zemlya*) centred on Berestye (*Brest*, 1019AD), the main toll station and fortress between Turov, Volhynia, Mazovia and Yatvingia. Between

1969 and 1981, P. F. Lysenko excavated 1,8k m² of the waterlogged *Hospital Island* citadel (4 - 7 m deep), uncovering 12 layers of pine log-corduroy streets (eleventh to thirteenth centuries) and over 220 log cabins preserved up to 12 logs (1,8 m) high with doorways, windows and oak roof-shingles [16]. Berestye covered 34 ha (7,5k inhabitants) at 1200AD, guarded to the north by Kamenets (1276AD, where Volhynian Prince Vladimir Vasilkovich and the builder Aleksa raised the 30 m brick Tower of Kamenets or *Belaya Vezha*, 3k inhabitants by 1300AD), Kobrin (2k) and Zditov (1,65k) [16].

Black Ruthenia and the Upper Sozh

Along the Upper Neman river, Slavic settlement by Dregovichi, Volhynians and Krivichi in Yotvingian and Lithuanian country created the principalities of Black Ruthenia (*Chornaya Rus*) [9][17][18]. The Principality of Grodno (*Goroden*, 1128-1201AD, 53,69°N, 23,83°E) was ruled by Prince Vsevolodko Davydovich (d. 1141AD, husband of Vladimir Monomakh's daughter Agafia) and his sons Boris and Gleb; excavated by J. Jodkowski and N. N. Voronin (1954), Grodno (35 ha, 8,5k inhabitants) developed its own style of brick and stone building seen in the Kalozha Church of SS Boris and Gleb (c. 1180s), the Lower Church and the Princely Palace on *Staraya Gora*, whose walls are inlaid with polished granite boulders and glazed ceramic crosses [17][18]. The Principality of Novogrudok (1201-1241AD, 53,6°N, 25,82°E) stood on the high hills of *Zamkovaya Gora* and *Maly Zamok* (323 m elevation), where F. D. Gurevich's excavations (1956-1985) uncovered a rich twelfth- and thirteenth-century boyar and goldsmith quarter on the *Okolny Gorod* with two-storey timber houses, glass windows, frescoed plaster walls, Byzantine amphorae, Syrian enamelled glass goblets and gold and silver casting moulds (32 ha in all, 7,5k inhabitants at 1200AD, rising to 9k by 1300AD as Mindaugas's early capital) [9]. The Principality of Volkovysk (1201-1241AD, 53,16°N, 24,45°E) covered the fortified hills of *Shvedskaya Gora*, *Zamchishche* and *Muravelnik* (24 ha, 5k inhabitants according to Ya. G. Zverugo), alongside Slonim (*Uslonim*, 12 ha, 3,5k), Turiisk-on-Neman (1,6k) and Zditovo-on-Molchad (1,5k) [18].

In eastern Belarus, the Principality of Mstislavl (1181-1198AD, 54,02°N, 31,73°E) and the Upper Dnieper-Sozh valley centred on Mstislavl, founded in 1135AD by Smolensk Prince Rostislav Mstislavich (d. 1167AD) on the Vikhra river and named after his father Mstislav the Great. Excavations by L. V. Alekseev and M. A. Tkachev on *Zamkovaya Gora* (26 ha in all, 5k inhabitants at 1200AD) recovered a birch-bark fragment, a princely seal and twelfth-century workshop layers. Together with Krichev (1136AD, 3k), Propoisk (*Prupoi*, 1136AD, 2,8k), Rechitsa (2,5k), Chechersk (1159AD, 2k), Luchin (1,85k), Shklov (1,75k), Mogilev (1,95k) and Gomel (*Homiy*, 1142AD, 30 ha, 6,5k inhabitants on the lower Sozh), this eastern Belarusian strip held 130k inhabitants in 1200AD, counted separately from the 340k inhabitants of eastern Smolensk east of 31°E in European Russia [19].

Principality Group	800AD Pop.	1000AD Pop.	1100AD Pop.	1200AD Peak	1250AD Low	1300AD Pop.	Fortified Towns
Polotsk, Vitebsk, Jersika & Koknese	135k	180k	230k	285k	260k	270k	46 (5 >10 ha)
Minsk, Drutsk & Borisov	65k	90k	115k	145k	130k	138k	34 (4 >10 ha)
Turov, Pinsk, Kletsk, Dubrovytsia & Berestye	130k	170k	215k	270k	195k	205k	48 (7 >10 ha)
Black Ruthenia: Grodno, Novogrudok & Volkovysk	70k	95k	120k	150k	145k	155k	32 (4 >10 ha)

Mstislavl, W. Smolensk & Upper Sozh (west of 31°E)	60k	85k	110k	130k	90k	92k	28 (3 >10 ha)
Total North-Western Rus (207,6k km^2)	460k	620k	790k	980k	820k	860k	188 (23 >10 ha)

Table 2. Sub-regional population estimates and excavated fortified town counts across the five river-basin groups of North-Western Rus from the Krivichi, Dregovichi and Radimichi tribal era (800AD) to the early Grand Duchy of Lithuania (1300AD).

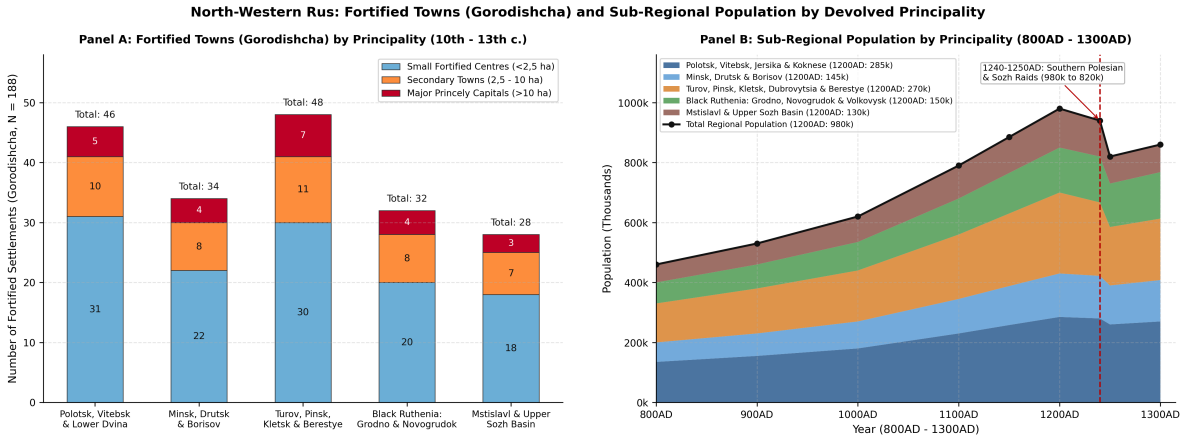


Figure 3. Left: distribution of the 188 excavated fortified settlements (gorodishcha) catalogued by Shtykhov, Lysenko, Zagorulsky and Kuza across the five Belarusian principality groups, distinguishing major capitals (>10 ha), secondary volost centres (2,5 - 10 ha) and small castles (<2,5 ha). Right: sub-regional population trajectories from 800AD to 1300AD, contrasting the 1239-1241AD losses in Turov-Pinsk and the Lower Sozh with the unstormed northern and western principalities.

The Mongol Invasion and Early Lithuanian Rule

The Campaigns of 1239-1241

Batu Khan's western campaign (1237-1241AD) produced a sharp divide between the exposed south and the sheltered north of Belarus [1][2][3]. Across southern Polesia and the Lower Sozh, destruction layers date directly to 1239-1241AD: in the autumn of 1239AD, Möngke's and Batu's forces marching against the Chernigov Principality stormed Gomel (*Homiy*), where excavations on the Sozh bluff uncovered burned timber ramparts, weapon caches and unburied skeletons in the 1239AD layer (6,5k in 1200AD falling to 3,5k in 1250AD). In the winter of 1240-1241AD, after the fall of Kiev, Mongol detachments riding west toward Volodymyr and Poland struck the southern Pripyat valley, sacking Turov (12k to 7k), Mozyr (3,5k to 2,1k), Bragin (2,8k to 1,5k), Dubrovytsia (3,5k to 2,2k) and Berestye (7,5k to 5,2k), where the *Hypatian Chronicle* records unburied corpses lying across the fields between Volodymyr and Berestye in early 1241AD [3][16]. Later raids led by the Mongol general Burundai (1258-1259AD) passed through Polesia and forced the southern frontier towns to pull down their timber walls.

Protected by the broad marshes of the Berezina-Pripyat lowlands and the upper Dnieper woods, Polotsk (25k in 1200AD, 22k in 1250AD and 23k in 1300AD), Vitebsk (9,5k, 8,8k and 9,5k), Minsk (9k, 8,2k and 8,8k), Drutsk (4,5k, 4,1k and 4,3k), Borisov (3,5k, 3,2k and 3,4k), Grodno (8,5k, 8k and 8,5k), Novogrudok (7,5k, 8,2k and 9k) and Volkovysk (5k, 4,7k and 5,1k) were never besieged or stormed during the 1237-1241AD invasion [1][7][9]. Although the collapse of Dnieper-Black Sea trade and Livonian crusader advances on the lower Dvina (*Jersika* and *Koknese* having fallen to Riga in 1208-1209AD) reduced Polotsk's river commerce, northern and western Belarus took in craftsmen and boyar families fleeing the ravaged Middle Dnieper.

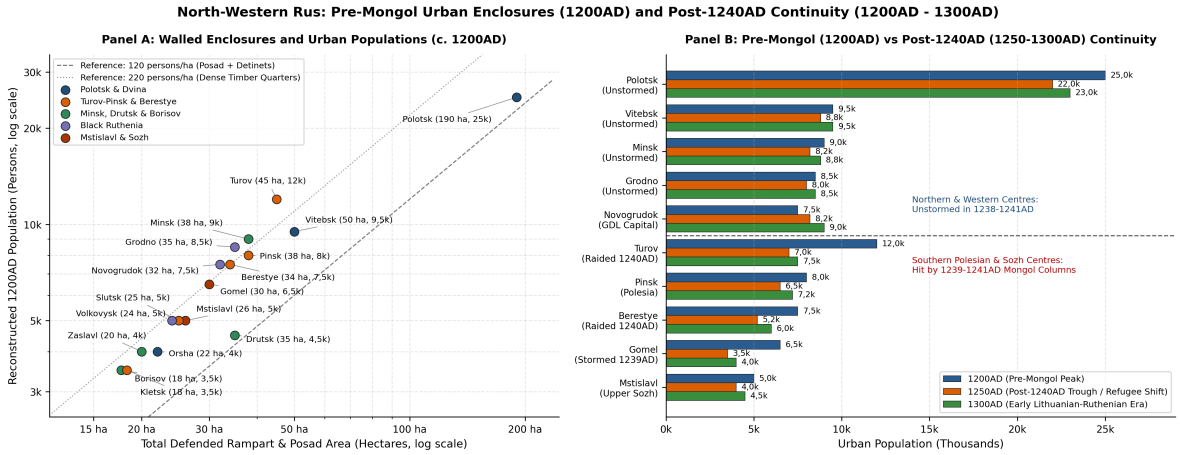


Figure 4. Left: walled rampart and suburban posad areas plotted against urban population in 1200AD, led by the 190 ha metropolis of Polotsk (25k), Turov (45 ha, 12k), Vitebsk (50 ha, 9,5k) and Minsk (38 ha, 9k). Right: urban population trajectories between 1200AD and 1300AD, contrasting the sacked southern towns of Gomel, Turov and Berestye with the continuity of Polotsk, Vitebsk and Minsk and the growth of Mindaugas's early capital at Novogrudok.

Early Lithuanian-Ruthenian Integration

In the decades after 1240AD, the unstormed towns and boyar assemblies (*veche*) of Black Ruthenia and the Polotsk Land formed the core of the early Grand Duchy of Lithuania (1240s - 1300AD) [9][10][23]. By the mid-1240s, Novogrudok, Grodno, Volkovysk and Slonim in Black Ruthenia served as the military and craft heartland of Mindaugas (d. 1263AD) and his Orthodox son Vaišvilkas (*Voishalk*, 1264-1267AD), who founded the Lavrishevo Monastery on the Neman near Novogrudok. Excavations in late thirteenth-century Novogrudok show steady growth across its boyar and craft quarters from 7,5k inhabitants in 1200AD to 8,2k in 1250AD and 9k in 1300AD [9].

In Polotsk and Vitebsk, through dynastic marriages and *veche* charters guaranteeing traditional local law (*starina*), Mindaugas's nephew Tautvilas (*Tovtivil*, Prince of Polotsk c. 1252-1263AD), Gerdenis (1264-1267AD, who signed the 1264AD Polotsk-Vitebsk trade treaty with Riga and Gotland) and Prince Konstantin Bezrukii (c. 1270s-1290s) kept Polotsk (23k in 1300AD) and Vitebsk (9,5k in 1300AD) as self-governing principalities allied with Lithuania against the Livonian Order and the Golden Horde [1][7][10]. Across North-Western Rus as a whole, total population dipped by 16,3% from its 1200AD pre-Mongol peak of 980k to 820k in 1250AD, before recovering to 860k (range 720k - 1M, 4,14 per km^2) by 1300AD (408k in Polotsk, Vitebsk, Minsk, Drutsk and Borisov; 205k in Turov, Pinsk, Kletsk, Dubrovytsia and Berestye; 155k in Black Ruthenia; and 92k in Mstislavl and the Upper Sozh), forming the heartland of the Lithuanian-Ruthenian state at the close of the thirteenth century [10][23].

Sources

1. Shtykhov, Georgiy Vasilevich. *Goroda Polotskoy Zemli (IX-XIII Vv.)*. Navuka i Tekhnika, 1992, https://search.worldcat.org/search?q=Goroda+Polotskoy+zemli+IX+XIII+vv+Shtykhov+1992.
2. Zagorulsky, Eduard Mikhailovich. *Arkheologiya Belarusi*. BGU, 2001, https://search.worldcat.org/search?q=Arkheologiya+Belarusi+Zagorulsky+2001.
3. Lysenko, Petr Fedorovich. *Drevniy Turov*. Belorusskaya Nauka, 2004, https://search.worldcat.org/search?q=Drevniy+Turov+Lysenko+2004.
4. Charniauski, Mikhail Mikhailavich, and Uladzimir Fedaravich Isaenko. *Arkhealogiya Belarusi, Tom 1: Kamenny i Bronzavy Vyaki*. Belaruskaya Navuka, 1997, https://search.worldcat.org/search?q=Arkhealogiya+Belarusi+Tom+Kamenny+bronzavy+vyaki+Charniauski+1997.
5. Shadyra, Vadzim Iosifavich, and Valiantsina Siargeeuna Viargey. *Arkhealogiya Belarusi, Tom 2: Zhaliezny Vek i Rannyae Syarednyavechcha*. Belaruskaya Navuka, 1999, https://search.worldcat.org/search?q=Arkhealogiya+Belarusi+Tom+Zhaliezny+vek+rannyae+syarednyavechcha+Shadyra+1999.
6. Sedov, Valentin Vasilevich. *Slavyane Verkhnego Podneprovya i Podvinya*. Nauka, 1970, https://search.worldcat.org/search?q=Slavyane+Verkhnego+Podneprovya+Podvinya+Sedov+1970.
7. Tarasov, Sergey Vasilevich. *Polatsk IX-XVII St.: Gistarychnaya Tapagrafiya i Demagrafiya*. Belaruskaya Navuka, 1998, https://search.worldcat.org/search?q=Polatsk+IX+XVII+st+Gistarychnaya+tapagrafiya+demagrafiya+Tarasov+1998.
8. Kuza, Andrey Vasilevich. *Malye Goroda Drevney Rusi*. Nauka, 1989, https://search.worldcat.org/search?q=Malye+goroda+Drevney+Rusi+Kuza+1989.
9. Gurevich, Frida Davydovna. *Drevniy Novogrudok: Posad i Okolny Gorod*. Nauka, 1981, https://search.worldcat.org/search?q=Drevniy+Novogrudok+Posad+okolny+gorod+Gurevich+1981.
10. Lowmianski, Henryk. *Studia Nad Dziejami Wielkiego Ksiestwa Litewskiego*. Wydawnictwo Naukowe UAM, 1983, https://search.worldcat.org/search?q=Studia+nad+dziejami+Wielkiego+Ksiestwa+Litewskiego+Lowmianski+1983.
11. Shtykhov, Georgiy Vasilevich. *Drevniy Polotsk IX-XIII Vv.* Nauka i Tekhnika, 1975, https://search.worldcat.org/search?q=Drevniy+Polotsk+IX+XIII+vv+Shtykhov+1975.
12. Alekseev, Leonid Vasilevich. *Polotskaya Zemlya: Ocherki Istorii Severnoy Belorussii v IX-XIII Vv.* Nauka, 1966, https://search.worldcat.org/search?q=Polotskaya+zemlya+Ocherki+istorii+severnoy+Belorussii+IX+XIII+Alekseev+1966.
13. Mugurevics, Evalds. *Vostochnaya Latviya i Sosednie Zemli v x-XIII Vv.* Zinatne, 1965, https://search.worldcat.org/search?q=Vostochnaya+Latviya+sosednie+zemli+XIII+vv+Mugurevics+1965.
14. Zagorulsky, Eduard Mikhailovich. *Vozniknovenie Minska*. Izdatelstvo BGU, 1982, https://search.worldcat.org/search?q=Vozniknovenie+Minska+Zagorulsky+1982.
15. Lysenko, Petr Fedorovich. *Goroda Turovskoy Zemli*. Nauka i Tekhnika, 1974, https://search.worldcat.org/search?q=Goroda+Turovskoy+zemli+Lysenko+1974.
16. Lysenko, Petr Fedorovich. *Berestye*. Nauka i Tekhnika, 1985, https://search.worldcat.org/search?q=Berestye+Lysenko+1985.
17. Voronin, Nikolay Nikolaevich. *Drevnee Grodno (Po Materialam Arkheologicheskikh Raskopok 1932-1949 Gg.)*. Izdatelstvo AN SSSR, 1954, https://search.worldcat.org/search?q=Drevnee+Grodno+materialam+arkheologicheskikh+raskopok+1932+1949+gg+Voronin+1954.
18. Zverugo, Yaroslav Genrikhovich. *Verkhnee Ponemanye v IX-XIII Vv.* Navuka i Tekhnika, 1989, https://search.worldcat.org/search?q=Verkhnee+Ponemanye+IX+XIII+vv+Zverugo+1989.
19. Alekseev, Leonid Vasilevich. *Smolenskaya Zemlya v IX-XIII Vv.: Ocherki Istorii Smolenshchiny i Vostochnoy Belorussii*. Nauka, 1980, https://search.worldcat.org/search?q=Smolenskaya+zemlya+IX+X

III+vv+Ocherki+istorii+Smolenshchiny+Alekseev+1980.

20. Sedov, Valentin Vasilevich. *Vostochnye Slavyane v VI-XIII Vv.* Nauka, 1982, https://search.worldcat.org/search?q=Vostochnye+slavyane+VI+XIII+vv+Sedov+1982.
21. Mitrofanov, Aleksey Grigorevich. *Zhelezny Vek Sredney Belorussii (VII v. Do n.e. - VIII v. N.e.)*. Nauka i Tekhnika, 1978, https://search.worldcat.org/search?q=Zhelezny+vek+Sredney+Belorussii+VII+VIII+Mitrofanov+1978.
22. Pobol, Leonid Davydovich. *Arkheologicheskie Pamyatniki Belorussii: Zhelezny Vek*. Nauka i Tekhnika, 1983, https://search.worldcat.org/search?q=Arkheologicheskie+pamyatniki+Belorussii+Zhelezny+vek+Pobol+1983.
23. Ochmanski, Jerzy. *Historia Litwy*. Ossolineum, 1982, https://search.worldcat.org/search?q=Historia+Litwy+Ochmanski+1982.